

ORIGINAL ARTICLE

Variation in Vasoactive Treatment Selection for Cardiogenic Shock: Insights From the Critical Care Cardiology Trials Network (CCCTN)

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BACKGROUND: The paucity of data to guide selection of specific vasoactive agents in patients with cardiogenic shock (CS) may lead to variability in practice patterns. The level of variability and specific factors that are associated with the use of vasoactive medications and inodilators have not been previously described.

METHODS: The CCCTN (Critical Care Cardiology Trials Network) is an international, multicenter network of cardiac intensive care units (CICUs) coordinated by the Thrombolysis in Myocardial Infarction Study Group. This analysis included CICU admissions for CS from 2019 to 2023. Variation in the use of inodilator treatment (dobutamine/milrinone) was assessed with multivariable mixed-effects logistic modeling.

RESULTS: A total of 3282 admissions from 37 CICUs comprised the analysis cohort. The use of vasoactive medications, including inodilator treatment, varied substantially across institutions. Patient-level variables associated with greater use of inodilators included history of heart failure (odds ratio, 1.98 [95% CI, 1.61–2.44]), biventricular failure (1.59 [95% CI, 1.27–2.00]), Society of Cardiovascular Angiography and Interventions stage D (1.34 [95% CI, 1.07–1.68]), valvular disease (1.34 [95% CI, 1.03–1.74]), and male sex (1.23 [95% CI, 1.02–1.49]). Variables associated with less inodilator use included cardiac arrest (0.33 [95% CI, 0.27–0.42]), right ventricular failure (0.50 [95% CI, 0.33–0.73]), Society of Cardiovascular Angiography and Interventions stage E (0.57 [95% CI, 0.41–0.81]), acute myocardial infarction-CS (0.71 [95% CI, 0.56–0.90]), peripheral arterial disease (0.73 [95% CI, 0.54–0.99]), older age (0.77 [95% CI, 0.72–0.83], per 10-year increase), and estimated glomerular filtration rate (0.96 [95% CI, 0.93–0.99], per 10 mL/min per 1.73 m² increase). No individual measurable institution-level factors (eg, transplant center) were associated with variability in inodilator use. In mixed-effects logistic modeling, 45.7% of variation in inodilator use was attributed to patient-level factors and 22.7% to the random effect of individual CICU centers. Similarly, 35.3% of variation in the use of dobutamine versus milrinone was attributed to patient-level factors and 32.6% to the random effect of individual CICU centers.

CONCLUSIONS: There is significant variation in vasoactive treatment and inodilator use in CS. Variation in inodilator use was associated with patient-level factors and with substantial individual CICU practice variation. Such variability underscores the need for additional high-quality evidence to guide vasoactive treatment strategies in CS.

Key Words: angiography ■ critical care ■ heart failure ■ odds ratio ■ shock

Editorial by Ortega-Hernández et al

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCHEARTFAILURE.125.013778>.

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Circulation: Heart Failure is available at www.ahajournals.org/journal/circheartfailure

WHAT IS NEW?

- This study quantifies the substantial variability in vasoactive medication selection for the treatment of cardiogenic shock across a large cohort of international cardiac intensive care units.
- While patient factors like heart failure history, shock severity, and ventricular predominance of shock influence inodilator use, over 20% of treatment variability is explained solely by which hospital a patient is admitted to, independent of measurable patient or institutional characteristics.
- The choice between dobutamine and milrinone shows even greater institutional variation, suggesting treatment selection is heavily driven by local practice culture rather than evidence-based criteria.

WHAT ARE THE CLINICAL IMPLICATIONS?

- The marked practice variability observed, independent of patient characteristics or institutional resources, highlights the urgent need for randomized trials comparing vasoactive treatment strategies in cardiogenic shock to inform standardized, evidence-based care.
- Clinicians should recognize that current treatment decisions may reflect local convention rather than proven superiority of 1 treatment option.
- These findings support ongoing efforts to develop clinical trials in cardiogenic shock and underscore opportunities for quality improvement initiatives to reduce unwarranted variation and potential care disparities in cardiogenic shock.

Nonstandard Abbreviations and Acronyms

AMI-CS	acute myocardial infarction-cardiogenic shock
CCCTN	Critical Care Cardiology Trials Network
CICU	cardiac intensive care unit
CS	cardiogenic shock
HF	heart failure
ICU	intensive care unit
MCS	mechanical circulatory support
PAD	peripheral arterial disease
SCAI	Society of Cardiovascular Angiography and Interventions

Cardiogenic shock (CS) is a complex syndrome characterized by low cardiac output leading to end-organ hypoperfusion, which is associated with significant morbidity and mortality.^{1–5} Vasoactive pharmacotherapy is typically the first line of treatment for CS and includes medications with diverse mechanisms of action ranging from pure vasopressors, inopressors, inodilators, to pure vasodilators.^{5,6} However, there is a lack of robust data

comparing the effectiveness of strategies to guide treatment selection.⁵ For example, inodilator therapy (dobutamine/milrinone) is commonly used as a treatment for patients with CS targeting both increases in cardiac contractility and peripheral vasodilation to increase stroke volume.⁶ However, the decision to implement an inodilator treatment strategy or which individual inodilator to use is not well understood. The clinical trials comparing vasoactive therapies for CS are limited in size, scope, population, diversity, and outcomes.^{5,7–11} As a result, professional society scientific statements and guidelines highlight this lack of high-quality data and the presence of uncertainty, avoiding firm recommendations for first-line vasoactive agents while suggesting consideration of patient-level factors (eg, blood pressure, arrhythmias) along with physician preferences and institution-level factors (eg, institutional familiarity and medication availability) when choosing vasoactive treatments for CS.^{12–14}

Substantial variation between institutions has been identified for multiple CS interventions within the modern cardiac intensive care unit (CICU).^{15–18} Overall, variation in clinical care can reflect utilization of precision medicine to fit treatment to the specific patient and pathophysiology (patient-level factors). However, variation in clinical practice can also be a sign of differences in local culture, clinical training, experience, provider preferences, and resource availability (institution-level factors) and can be associated with worse health outcomes and greater health care disparities.^{19–22} Therefore, identifying the contribution of specific factors and quantifying their importance in this variability is crucial to discovering disparities in care and identifying gaps in current knowledge, as well as constructing strategies to standardize best practices for quality improvement.²³

In this context, current vasoactive practice patterns and factors impacting the selection of inodilator therapy in CS have not been well described. We utilized an international multicenter registry of CICU care to (1) quantify the variability in vasoactive treatment selection in CS; (2) quantify the variability of inodilator use in CS; and (3) identify patient-level and institutional-level factors that are associated with variation in inodilator use.

METHODS

Study Population

The CCCTN (Critical Care Cardiology Trials Network) is an investigator-initiated research network of CICUs across the United States, United Kingdom, and Canada, coordinated by the Thrombolysis in Myocardial Infarction Study Group (Boston, MA), with methods and cohort profile that have been described in detail.^{24,25} The registry includes clinical data on all consecutive CICU admissions regardless of diagnosis during annual minimum 2-month collection periods. The CCCTN Registry protocol and waiver of informed consent were approved by the institutional review boards at each of the participating centers. We encourage parties interested in collaboration and data

sharing to contact the corresponding author directly for further discussions.

This analysis includes all CS admissions from September 2019 to August 2023 for which a vasoactive agent was used within the first 24 hours of CICU admission. The study cohort could include multiple admissions from the same patient. To avoid possible confounding from the COVID-19 pandemic, patients with a diagnosis of COVID-19 were excluded from all analyses. To improve the stability of site-level estimates and reliability of institutional comparisons, the analysis was restricted to institutions with CICUs that had at least 20 CS cases during the study period.

Assessment of Vasoactive Therapies in CS

Shock was defined as sustained hemodynamic impairment (systolic blood pressure <90 mm Hg or the need for a vasoactive agent) with evidence of end-organ hypoperfusion.²⁴ Shock cause was categorized by site investigator and included in the analysis cohort if cardiogenic. Patients with alternative etiologies of shock (eg, hypovolemic, distributive) and patients with mixed shock were excluded from the analysis. Site investigators also reported CS characteristics including primary etiologic categories (acute myocardial infarction-related CS [AMI-CS], heart failure [HF]-related CS, and nonmyocardial [secondary] CS [eg, cardiac tamponade and severe valvular disease]), Society of Cardiovascular Angiography and Interventions (SCAI) stage, and ventricular predominance according to the Shock Academic Research Consortium phenotypic classification recommendations.²⁶ Continuous intravenous vasoactive agents used within the first 24 hours of CICU admission were captured in the electronic case record form, and included dobutamine, milrinone, dopamine, epinephrine, norepinephrine, vasopressin, and phenylephrine. Dosing information for each vasoactive agent was evaluated at the 4- and 24-hour time points from CICU admission and summarized using the Vasoactive-Inotropic Score.²⁷ Inodilator use was defined as the use of either dobutamine, milrinone or both, within the first 24 hours of CICU admission. Admissions in which both dobutamine and milrinone were prescribed within the first 24 hours of CICU admission, were included in the study cohort and initial comparative analyses between inotrope and no inotrope use. However, these admissions were excluded from the comparative analyses between dobutamine and milrinone. Temporary mechanical circulatory support (MCS) devices placed either at an outside hospital before transfer or within the first 36 hours of CICU admission were evaluated as a patient-level variable.¹⁷

Statistical Methods

Descriptive statistics utilized the median and interquartile range for the display of continuous variables and the number (n) and percentage for categorical variables. Comparisons were made with the Wilcoxon rank-sum test for continuous variables and the χ^2 test for categorical variables. The dot plot figure was created by transforming the data to have the number of dots represent the number of each vasoactive agent used per 500 admissions with CS. Multivariable mixed-effects logistic modeling was used to identify patient- and institution-level factors relating to variability in inodilator usage. Separately, multivariable mixed-effects logistic modeling was also used to compare dobutamine versus milrinone as the selected inodilator. Variables included in the multivariable mixed-effects logistic model were

separated into patient-level and institution-level fixed covariates, along with the individual CICU center as a random intercept to account for baseline between-center differences in treatment practice. Patient-level covariates were selected based on clinical relevance and prior literature regarding factors known or hypothesized to influence vasoactive treatment decisions in CS, and included demographic characteristics (age and sex), relevant comorbidities and history (HF, pulmonary hypertension, valvular heart disease, estimated glomerular filtration rate, peripheral arterial disease [PAD], and ventricular arrhythmias), and shock characteristics (cardiac arrest, MCS, AMI-CS, initial SCAI stage [Stage B or C as reference], and ventricular predominance). Institution-level covariates included CS cases/month, country, hospital location, number of hospital beds, number of CICU beds, and heart transplant capabilities. To assess the relative contribution of each component to the outcome of inodilator usage, variance partitioning was performed by extracting the variance estimates for the patient-level fixed effects, institution-level fixed effects, the random intercept for CICU center, and residuals from the mixed-effects logistic regression model.^{16,28} The percentage of total variance attributable to each component was then calculated as the proportion of that component's variance relative to the sum of all variance components. These variance decomposition estimates are specific to the models fit in this sample and should not be interpreted as generalizable population-level parameters. Temporal trends in vasoactive medication use were evaluated by the Cochran-Armitage trend test limited to intensive care unit (ICU) centers that participated in all of the study campaigns. Sensitivity analyses were conducted in 2 subpopulations, excluding transfer admissions and admissions from the single center within the United Kingdom, to evaluate if treatment variability persists even within United States and Canadian populations and nontransfer CS patients. A sensitivity analysis was also performed excluding PAD as a variable, given the absence of overt biological plausibility. Finally, a sensitivity analysis was conducted in patients who had placement of a pulmonary artery catheter and availability of invasive hemodynamic data. To achieve model convergence in the reduced sample size of admissions with available pulmonary artery catheter data, site-level covariates were excluded from these models. A *P* value of <0.05 was considered statistically significant. Analyses were performed using SAS, version 9.4 (SAS Institute, Inc, Cary, NC).

RESULTS

A total of 3282 admissions with CS treated with vasoactive therapy in the first 24 hours met the criteria for inclusion in the analysis cohort. Within this cohort, 2159 (65.8%) admissions were managed with an inodilator (milrinone and dobutamine) in the first 24 hours.

Characteristics of the Population

The overall population had an average age of 66 years, 32% were women, and 57% were White (Table 1). Notable features of this population include the presence of medical comorbidities, including type 2 diabetes (37%), chronic kidney disease (28%), coronary artery disease (36%), atrial fibrillation (32%), and HF (56%). Overall, 25% of

Table 1. Baseline Characteristics

	Overall (n=3282)	Inodilator (n=2159)	No inodilator (n=1123)	P value
Demographics				
Age, y; med. (IQR)	66 (56–74)	64 (54–72)	69 (59–78)	<0.0001
Sex, female; n (%)	1063 (32)	639 (30)	424 (38)	<0.0001
BMI, kg/m ² ; med. (IQR)	27 (24–32)	27 (23–32)	28 (24–33)	0.0006
Race, n (%)				<0.0001
White	1872 (57)	1136 (53)	736 (66)	
Black	644 (20)	505 (23)	139 (12)	
Other/Unknown	766 (23)	518 (24)	248 (22)	
History, n (%)				
Hypertension	1865 (57)	1168 (54)	697 (62)	<0.0001
Type 2 diabetes	1207 (37)	803 (37)	404 (36)	0.49
Chronic kidney disease	929 (28)	637 (30)	292 (26)	0.035
Dialysis	129 (4)	65 (3)	64 (6)	<0.0001
Pulmonary disease	419 (13)	254 (12)	165 (15)	0.017
Active cancer	188 (6)	100 (5)	88 (8)	0.0002
Cardiovascular history, n (%)				
Coronary artery disease	1176 (36)	771 (36)	405 (36)	0.84
Cerebrovascular disease	248 (8)	162 (8)	86 (8)	0.87
Peripheral arterial disease	276 (8)	161 (8)	115 (10)	0.006
Heart transplant	51 (2)	42 (2)	9 (1)	0.012
Atrial fibrillation	1054 (32)	728 (34)	326 (29)	0.006
Ventricular arrhythmias	268 (9)	203 (9)	65 (6)	0.0003
Pulmonary hypertension	243 (7)	170 (8)	73 (7)	0.15
Heart failure	1820 (56)	1389 (64)	431 (38)	<0.0001
HFrEF	1383 (78)	1146 (85)	237 (58)	<0.0001

Demographic descriptive statistics for the overall study population, patients that used inodilators, and patients that did not use inodilators for the treatment of CS in the CCCTN. Descriptive statistics use median and IQR for continuous variable display as well as number (n) and percentage for categorical variables. Comparisons were made with Wilcoxon rank-sum test for continuous variables and χ^2 test for categorical variables. BMI indicates body mass index; CCCTN, Critical Care Cardiology Trials Network; CS, cardiogenic shock; HFrEF, heart failure with reduced ejection fraction; and IQR, interquartile range.

admissions were AMI-CS, 63% were HF-related CS, and 12% were secondary CS. Admissions varied according to the pattern of ventricular predominance (left ventricular: 53%, right ventricular: 6%, biventricular: 29%) and SCAI Shock Stage (B: 8%, C: 57%, D: 26%, and E: 8%; Table 2). Within the CICU, 48% of patients were mechanically ventilated, 59% had a pulmonary arterial catheter placed, and 37% had placement of temporary MCS.

Overall Vasoactive Use

Within the overall population, in the first 24 hours, patients received an average of 2 vasoactive agents, the most common of which was norepinephrine. However, the use of each vasoactive agent varied widely by CICU institution (Figure 1; Table 3). The specific vasoactive medication class of inodilators (milrinone or dobutamine) was used during the first 24 hours in the treatment of CS in 65.8% of all admissions that required vasoactive therapy with variation across institutions

(interhospital range, 30.8%–91.1%; Figure 2). Within the inodilator class, considerable variability was noted for the use of dobutamine (interhospital range, 10%–81.6%) and milrinone (interhospital range, 6.3%–71.2%).

Inodilator Treatment Variability

Demographic and clinical factors differed between admissions that were managed with inodilators versus without inodilators (Tables 1 and 2). In particular, admissions managed with inodilators were generally younger and were more likely to be men, Black or of other races, and have a history of HF. They were less likely to have experienced a cardiac arrest or be admitted with AMI-CS. Admissions managed with inodilators were also more likely to have a pulmonary artery catheter, have biventricular failure, and have SCAI Stage C or D, and less likely to use mechanical ventilation and have SCAI Stage E compared with admissions that were not managed

Table 2. Clinical and ICU Center Characteristics

	Overall (n=3282)	Inodilator (n=2159)	No inodilator (n=1123)	P value
Admission				
Cardiac arrest, n (%)	783 (24)	345 (16)	438 (39)	<0.0001
Lactate, mmol/L; med. (IQR)	2.7 (1.6–5.2)	2.5 (1.5–4.6)	3.3 (1.7–6.7)	<0.0001
SOFA score; med. (IQR)	7 (5–10)	7 (5–9)	8 (6–11)	<0.0001
eGFR, mL/min per 1.73 m ² ; med. (IQR)	45 (27–65)	44 (27–64)	46 (29–66)	0.118
Shock onset, n (%)				
				<0.0001
HF-CS, n (%)	2056 (63)	1573 (73)	483 (43)	
AMI-CS, n (%)	815 (25)	445 (21)	370 (33)	
Secondary/other CS, n (%)	411 (12.5)	141 (6.5)	270 (24)	
Failure anatomy				
				<0.0001
Left ventricular	1729 (53)	1171 (54)	558 (50)	
Right ventricular	199 (6)	106 (5)	93 (8)	
Biventricular	955 (29)	766 (36)	189 (17)	
Other	399 (12)	116 (5)	283 (25)	
SCAI shock stage				
				<0.0001
B	262 (8)	147 (7)	115 (10)	
C	1874 (57)	1285 (60)	589 (53)	
D	860 (26)	613 (28)	247 (22)	
E	273 (8)	112 (5)	161 (15)	
CICU therapies and monitoring				
No. of vasoactives, med. (IQR)	2 (1–3)	2 (1–3)	1 (1–2)	<0.0001
VIS score at 4 h, med. (IQR)	5.0 (2.5–14.0)	5.0 (2.5–10.8)	6.3 (2.0–19.3)	0.0002
VIS score at 24 h, med. (IQR)	5.0 (2.0–10.4)	5.0 (2.5–10.0)	3.0 (0.0–11.0)	<0.0001
RRT, n (%)	456 (14)	312 (15)	144 (13)	0.20
Mechanical ventilation, n (%)	1560 (48)	868 (40)	692 (62)	0.001
Pulmonary arterial line, n (%)	1921 (59)	1491 (69)	430 (38)	<0.0001
MCS, n (%)	1223 (37)	879 (41)	344 (31)	<0.0001
Center				
Cases, mo; med. (IQR)	11 (9–17)	11 (8–17)	11 (10–17)	0.004
Country, n (%)				
				0.006
United States	2812 (86)	1869 (87)	943 (84)	
Canada	411 (12.5)	245 (11)	166 (15)	
United Kingdom	59 (2)	45 (2)	14 (1)	
Location, n (%)				
				<0.0001
Urban	2718 (83)	1835 (85)	883 (79)	
Suburban	477 (15)	271 (13)	206 (18)	
Rural	87 (3)	53 (3)	34 (3)	
Hospital beds, med. (IQR)	793 (614–1000)	793 (614–900)	793 (600–1000)	0.001
CICU beds, med. (IQR)	14 (10–23)	16 (12–21)	14 (10–24)	<0.0001
Transplant center, n (%)	2576 (79)	1678 (78)	898 (80)	0.14

Clinical, and CICU center descriptive statistics for the overall study population, patients that used inodilators, and patients that did not use inodilators for the treatment of CS in the CCCTN. Descriptive statistics utilized median and IQR for continuous variable display as well as number (n) and percentage for categorical variables. Comparisons were made with Wilcoxon rank-sum test for continuous variables and χ^2 test for categorical variables. AMI-CS indicates acute myocardial infarction-cardiogenic shock; CCCTN, Critical Care Cardiology Trials Network; CICU, Cardiac Intensive Care Unit; CS, cardiogenic shock; HF-CS, heart failure cardiogenic shock; ICU, intensive care unit; IQR, interquartile range; MCS, temporary mechanical circulatory support; RRT, renal replacement therapy; SCAI, Society of Cardiovascular Angiography & Intervention; and SOFA, sequential organ failure assessment.

with inodilators. Admissions treated with inodilators had a lower 4-hour Vasoactive-Inotropic Score but a higher 24-hour Vasoactive-Inotropic Score compared with

admissions treated without inodilators. Finally, admissions treated with inodilators were more likely to be at an urban CICU with greater CICU capacity.

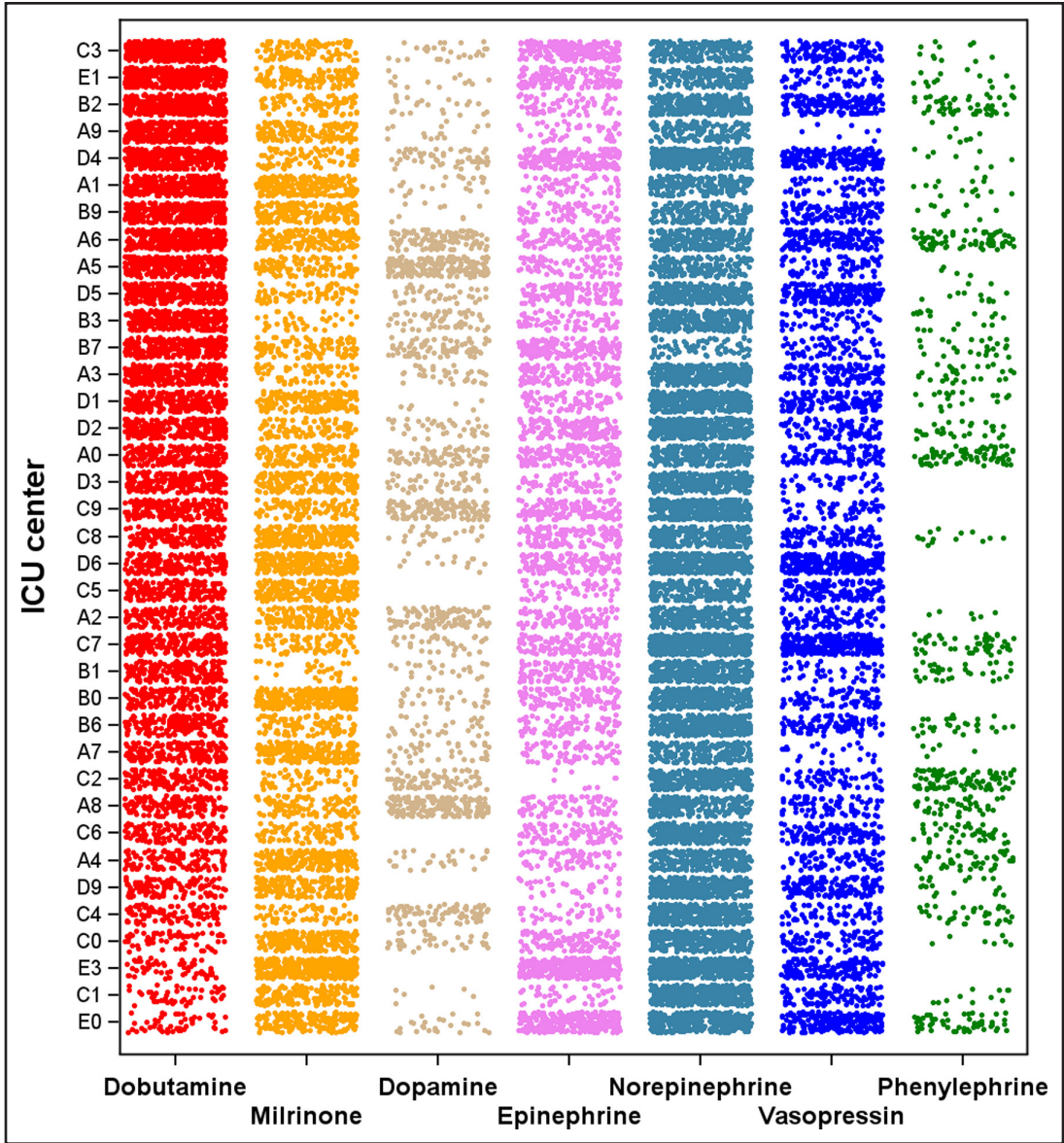


Figure 1. Vasoactive medication variability for the treatment of cardiogenic shock (CS). Dot plot demonstrating frequency of vasoactive agent use by institution in subjects with CS. The number of dots represents the number of uses of each vasoactive agent per 5 hundred patients in CS. Each individual y-axis tick mark represents a different institution within the CCCTN registry (Critical Care Cardiology Trials Network). ICU indicates intensive care unit.

In multivariable analysis, after adjustment for other covariates in the model, patient-level factors associated with greater inodilator use included HF history, biventricular failure, severe valvular disease, SCAI Stage D (SCAI B and C as reference), and male sex (Figure 3). Patient-level factors associated with less inodilator use included cardiac arrest, isolated right ventricular failure, SCAI Stage E (SCAI B and C as reference), AMI-CS, PAD history, older age, and higher estimated glomerular filtration rate at baseline. Notably, there was no association with temporary MCS use, history of ventricular arrhythmias, or history of pulmonary hypertension. Further, inodilator use was not independently associated with any of the measurable institution-level factors, including hospital

size, country, CS volume, number of CICU beds, and transplant center status. Overall, in this model, 45.7% of variability in inodilator use was explained by patient-level factors and 0.5% was explained by measurable institution-level factors (Figure 4). After accounting for the patient- and measurable institution-level fixed covariates, in this model, 22.7% of variability in inodilator use was explained by the random effect of individual CICU institutions (Figure 4).

Dobutamine Versus Milrinone

Within the cohort treated with inodilators, 28.8% of admissions were treated with milrinone alone, 59.3%

Table 3. Frequency of Vasoactive Agent Use in Cardiogenic Shock Across CICU Centers

	Median (%)	IQR (25–75)	Range
Dobutamine	37.7	30.2–61.9	10.0–81.6
Milrinone	29.9	17.9–36.4	6.3–71.2
Dopamine	5.8	2.9–11.5	0–37.2
Epinephrine	19.5	13.6–26.9	1.6–64.4
Norepinephrine	61.4	44.7–69.6	13.8–90.0
Vasopressin	19.6	13.8–28.6	1.5–52.9
Phenylephrine	2.9	1.4–7.3	0–17.5

Frequency of use of each individual vasoactive agents in admissions with cardiogenic shock by individual institution within the CCCTN during the first 24 h of CICU admission with the median percentage, IQR, and range from lowest frequency institution to highest frequency institution. CCCTN indicates Critical Care Cardiology Trials Network; CICU, cardiac intensive care unit; ICU, intensive care unit; and IQR, interquartile range.

with dobutamine alone, and 11.8% with both dobutamine and milrinone. Patient-level factors associated with greater use of dobutamine in comparison with milrinone included PAD, cardiac arrest, and older age (Figure 5). Patient-level factors associated with less dobutamine use in comparison with milrinone included history of HF, history of ventricular arrhythmias, and higher baseline estimated glomerular filtration rate. Milrinone use was more common in the UK center and in Canada compared with the United States. However, the selection of dobutamine versus milrinone was not associated with any other institution-level factors (Figure 5). Overall, in this model, 35.3% of variability in use of dobutamine versus milrinone was explained by patient-level factors and 1.3% of variability was explained by measurable institution-level factors (Figure 4). After taking patient- and measurable institution-level fixed covariates into account, in this model, 32.6% of variability in the use of dobutamine versus milrinone was explained by the random effect of individual CICU centers (Figure 4).

Temporal and Geographic Variation

When evaluated by individual year, there was a notable decrease in the use of dopamine and no trend was seen in the use of any other vasoactive medication over the 4-year study period (Table S1). In addition, when ICUs were categorized by geographic region within the United States, there was a difference in the frequency of each vasoactive medication used in CS treatment in an unadjusted analysis (Table S2).

Sensitivity Analyses

The mixed-effects logistic modeling was reanalyzed examining patient- and institution-level factors associated with inodilator use and comparison of dobutamine and milrinone treatment excluding the UK site (Tables S3 and S4), excluding patients that underwent institutional transfer

(Tables S5 and S6), and, given the absence of overt biological plausibility, excluding the PAD variable (Tables S7 and S8). Overall, these sensitivity analyses demonstrated results consistent with the primary analysis (Table S9). For patients with available invasive hemodynamic data (n=1674), in the comparison of inodilator use, higher systemic vascular resistance demonstrated a statistically significant relationship with increased inodilator use ([1-SD]; odds ratio, 1.38 (95% CI, 1.14–1.68); P=0.001), while cardiac index was not significantly associated ([1-SD]; odds ratio, 0.98 (95% CI, 0.84–1.16), P=0.87; Table S10). Similarly, for comparison of dobutamine versus milrinone use, higher systemic vascular resistance was associated with more milrinone use (odds ratio, 0.80 (95% CI, 0.66–0.98); P=0.03), while cardiac index was not significantly associated (odds ratio, 0.89 (95% CI, 0.74–1.08), P=0.25; Table S11). In these models including systemic vascular resistance and cardiac index, the direction and general magnitude of associations of other patient-level factors remained consistent with the primary analyses.

DISCUSSION

In this international multicenter registry of 37 CICUs including 3282 admissions, there was significant variability in the use of vasoactive treatment strategies for the management of CS across all classes of vasoactive medications. Within the inodilator class of vasoactive medications, there was significant variability in the frequency of use within CS and in the selection of dobutamine versus milrinone across institutions. Variability in inodilator use was only partially explained by patient-level factors but was minimally associated with measurable institution-level factors (eg, transplant center). Mixed-effects modeling demonstrated that individual CICU institutional clinical practice patterns contributed significantly to variability in inodilator use. These findings suggest that the role of local/regional culture and clinical practice variation may be a fundamental element of CS care and may reflect a lack of high-quality data and an opportunity for standardization of treatment practices.

Implications

Overall, vasoactive treatment of CS has significant variability across the CCCTN registry highlighted by the broad ranges of use of all vasoactive medications, including vasopressors, inopressors, and inodilators. Notably, for inopressors like norepinephrine, epinephrine, and dopamine there was variability in treatment frequency across ICUs within the CCCTN registry. Prior trials like the Comparison of Dopamine and Norepinephrine in the Treatment of Shock (SOAP-II) trial, the Catecholamine Study: A Comparison of Epinephrine and Norepinephrine in Critical Ill Patients (CAT) trial, and Study Comparing the Efficacy and Tolerability of Epinephrine and Norepinephrine in Cardiogenic

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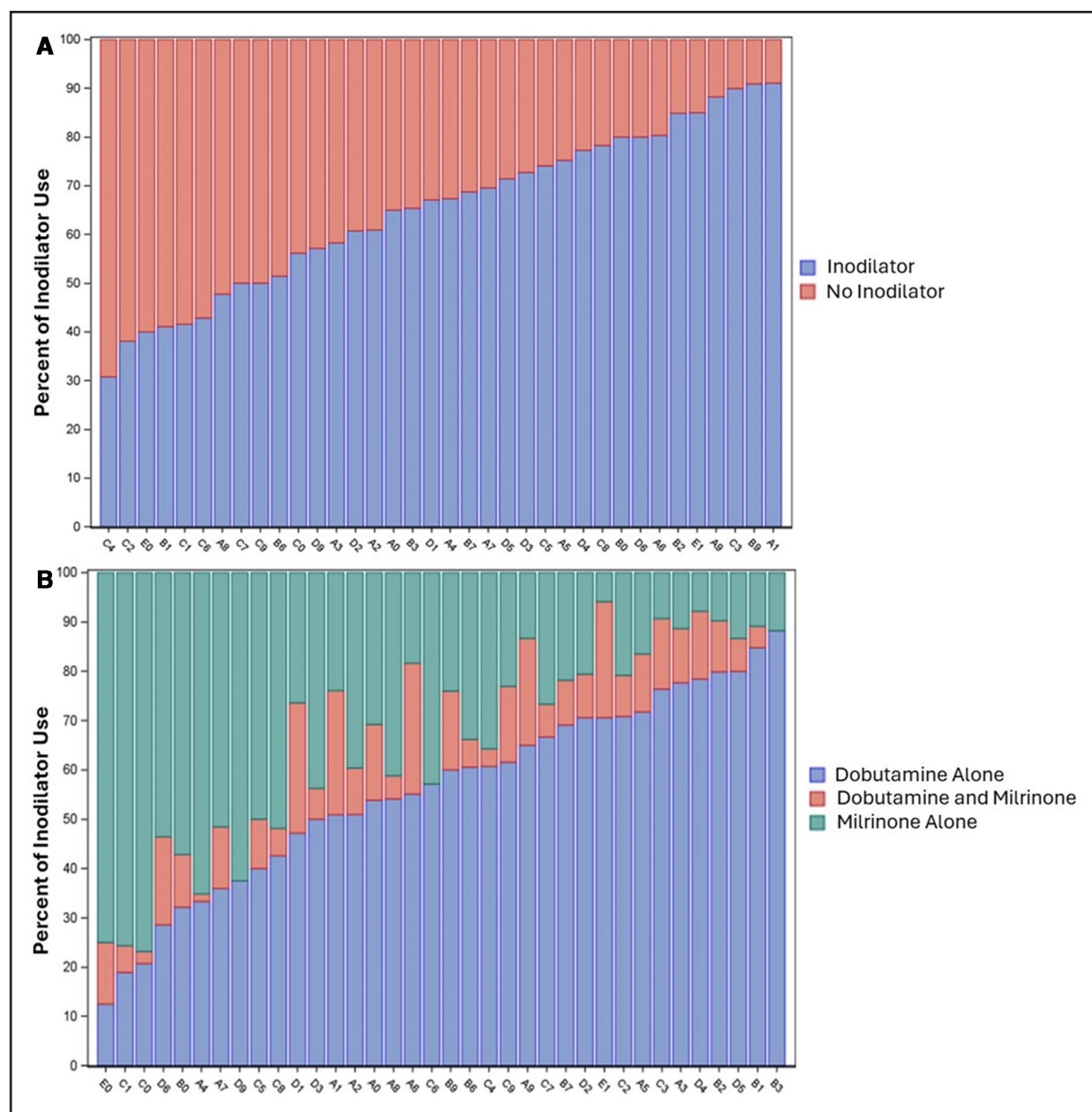


Figure 2. Rate of inodilator use by institution in subjects with cardiogenic shock.

A, Describes frequency of inodilator usage within the total patient cohort by institution. For patients that were treated with inodilator therapy.

B, Describes the frequency of use of individual inodilator therapy. Each individual x-axis tick mark represents a different institution within the CCTN registry (Critical Care Cardiology Trials Network).

Shock (Optima-CC) trial offer comparisons of these vasoactive medications in various populations (SOAP-II: 1679 shock patients, CAT: 280 shock patients, and Optima-CC: 57 AMI-CS patients) without a conclusive mortality benefit of 1 treatment option over another.^{7,9,10} The absence of definitive trial data has created an opportunity for this variability to exist without data to establish which treatment is superior and how this relates to different populations and disease states.

For the variability seen in the use of inodilators, there were numerous patient-level factors that impacted treatment choice. Vasomotor tone could be an important

variable in the decision-making of CS treatment selection, as evidenced by the association between higher systemic vascular resistance and increased inodilator use. In addition, physiological states commonly associated with elevated vasomotor tone (HF-related CS) were associated with increased inodilator therapy, while states associated with impaired vasomotor tone (postcardiac arrest syndrome, AMI-CS, and SCAI stage E) were associated with less inodilator therapy.²⁹ This potentially implicates vasomotor tone and its perceived impact on efficacy and tolerability of inodilator therapy as an important variable in the treatment selection. However, other patient-level

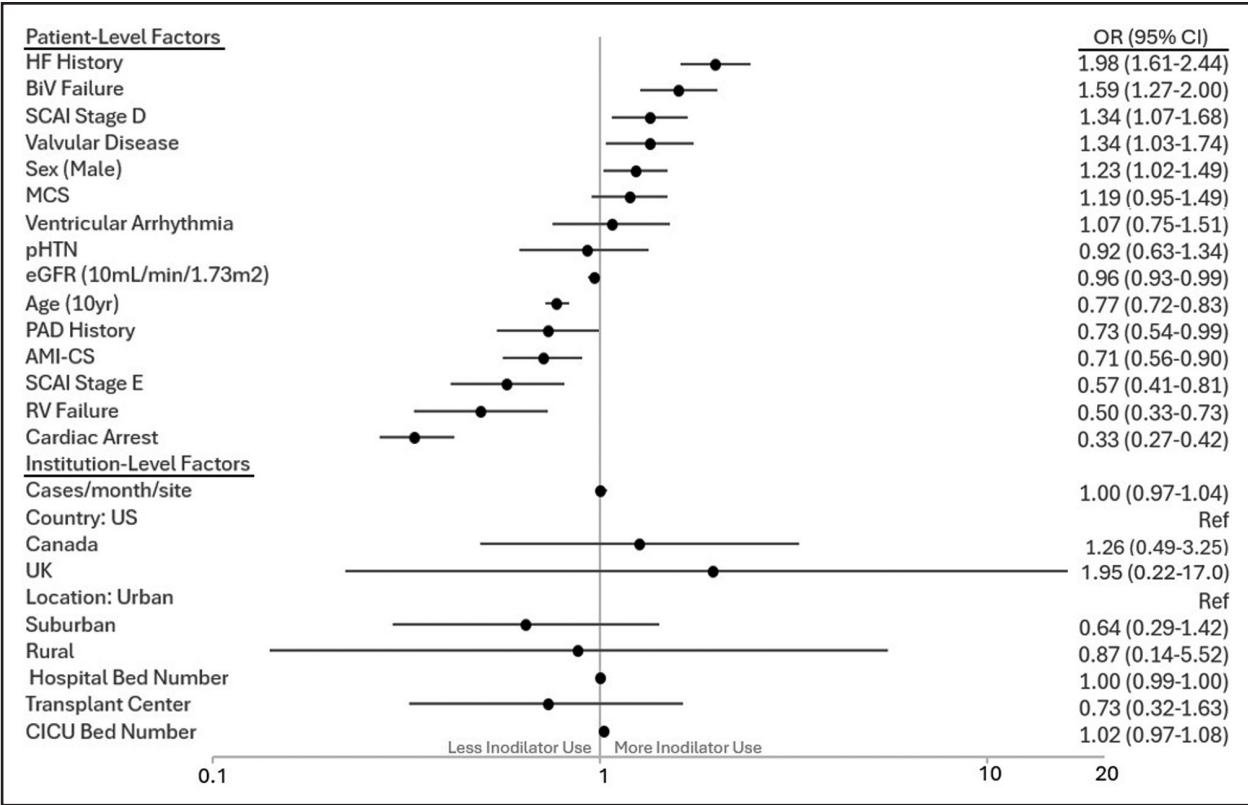


Figure 3. Patient- and institution-level factors associated with use of inodilator treatment for patients in cardiogenic shock. Mixed-effect logistic modeling demonstrating odds ratio (OR) and 95% CI graphically depicted in log-scale. AMI-CS indicates acute myocardial infarction cardiogenic shock; BiV, biventricular; CICU, cardiac intensive care unit; eGFR, estimated glomerular filtration rate; HF, heart failure; LV, left ventricular; MCS, mechanical circulatory support; PAD, peripheral artery disease; pHTN, pulmonary hypertension; RV, right ventricular; SCAI, Society of Cardiovascular Angiography & Interventions; UK, United Kingdom; and US, United States.

factors independently associated with differences in inodilator use, such as age and sex, highlight a need for standardized practice.

Importantly, even after taking patient- and individual measurable institution-level factors that could impact resource availability, training, experience, and expertise into account, a substantial proportion of the variability (>20%) in the use of inodilator therapy was related only to which CICU the patient happened to be admitted. The lack of association between certain institution-level characteristics, such as transplant center status or monthly CS case volume, suggests that institutional variation in clinical decision-making is influenced by factors beyond institutional capacity, experience, or expertise. Prior studies in the use of pulmonary artery catheter and temporary MCS within the CICU population and the use of vasoactive agents in the cardiac surgery population have corroborated similar themes to this study demonstrating significant variation in management of CS across institutions.^{16–18}

The observed variability within this study underscores the need for more robust evidence to inform standardized treatment guidelines, such as the ongoing Inotrope Versus Placebo Therapy for Cardiogenic Shock (CAPITAL DOREMI2) trial, a randomized clinical trial comparing inodilator treatment to placebo in patients with CS.³⁰

While individualized care is plausibly necessary for optimizing outcomes in complex syndromes like CS, excessive variability can arise from local culture and personal practice variation leaving patients vulnerable to disparities in care and outcomes. This study's findings suggest that the clarification and standardization of treatment protocols, guided by additional comprehensive multi-center data and randomized clinical trials, could reduce unwarranted variability without undermining the potential for personalized approaches.

Similarly, for the inodilator treatment decision of using dobutamine versus milrinone, there was substantial variation across institutions. However, compared with inodilator treatment overall, patient- and institution-level factors had a smaller impact on the overall variability, and fewer patient-level factors were associated with this variability. Importantly, the Dobutamine Compared with Milrinone (DOREMI) trial was published in the middle of the study cohort timeline and compared the use of dobutamine versus milrinone for patients with CS, finding no difference in the composite in-hospital cardiac, cerebrovascular, and renal outcome.⁸ As a pragmatic randomized controlled trial, this trial was a valuable starting point to guide treatment decisions in CS. We note that the variability in dobutamine and milrinone use may be in part driven by the clinical

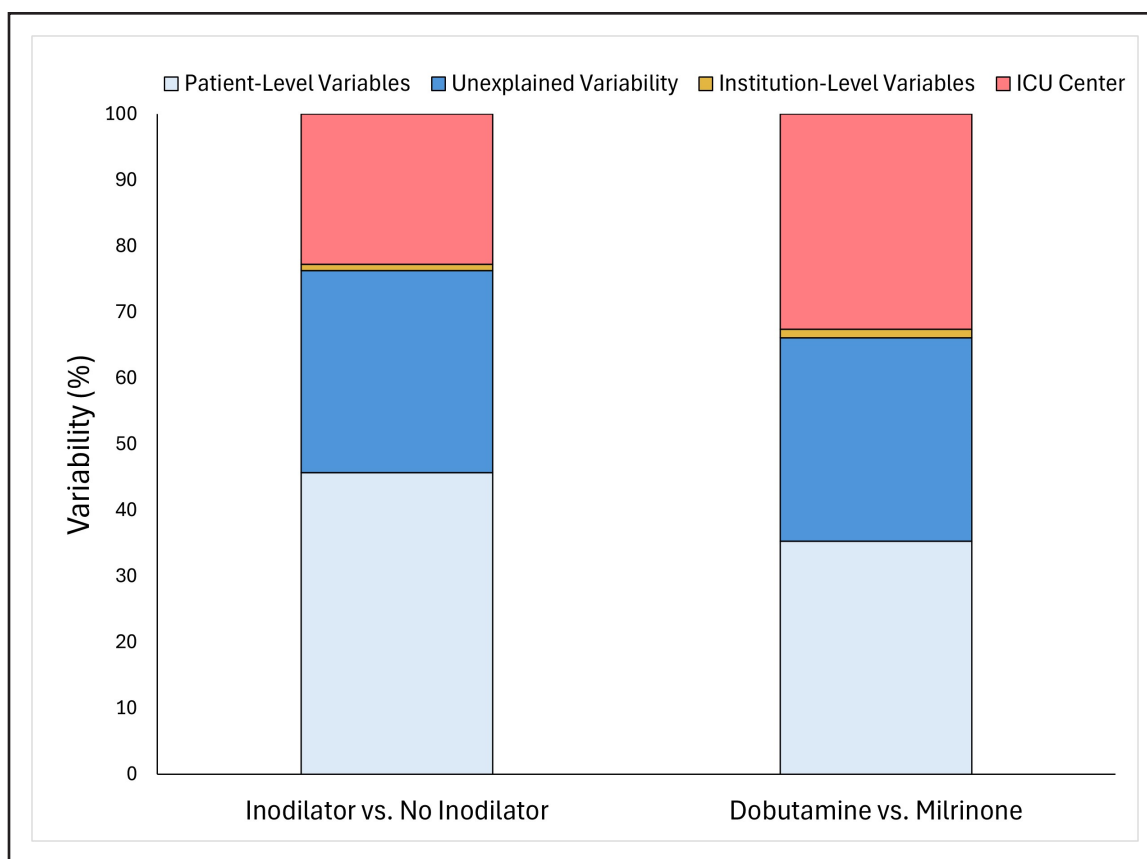


Figure 4. Factors associated with variability in inodilator use.

Multivariable mixed-effect logistic modeling describing the factors associated with variability in inodilator use and use of dobutamine vs milrinone. Each slice of the pie graphic represents the estimated percentage of variance that is associated with (1) patient-level factors, (2) institution-level factors, (3) unexplained variability, and (4) intensive care unit (ICU) center alone.

equipoise highlighted by this trial. However, there remains a vital opportunity for larger trials that take into account specific patient subpopulations with a physiological basis for possible differential outcomes (eg, right ventricular failure, pulmonary hypertension, and renal dysfunction) to potentially allow for more personalized approaches to inodilator therapy selection. In our population, a higher estimated glomerular filtration rate was associated with increased milrinone use, consistent with physiologically and clinically hypothesized subgroup benefits.³¹ However, right ventricular failure and history of pulmonary hypertension were not associated with increased milrinone use. Our work also identified that patients with a history of PAD and who suffered cardiac arrest were associated with increased dobutamine use, and patients with a history of HF or ventricular arrhythmia were associated with increased milrinone use, suggesting these subpopulations may be important to consider in future clinical trials.

In light of the potential impact of emerging trial data, our study describes that practice patterns have notably evolved during the 4-year study period. This variation could be secondary to evolving the data landscape with trials like SOAP-2, educational efforts, institutional protocolization, staffing models, medication availability/shortages, or

changing CICU subspecialty provider makeup.^{7,31} Similarly, geographic variation for vasoactive treatment within different regions of the United States as well as internationally highlights the impact of physician, institution, regional, cultural, and national practice patterns.¹⁵ These differences highlight the need for further study and standardization.

To address the variability in vasoactive treatment of patients with CS, we would recommend further understanding of both provider- and institution-level attitudes and opinions on the treatment of CS (eg, medication selection, escalation/deescalation sequencing, impact of hemodynamic monitoring and titration) using a mixed-method analysis.³² This additional study is needed to identify possible process interventions to facilitate evidence-informed care, align practice patterns across institutions, and reduce unwarranted variability. Finally, these data emphasize the need for randomized clinical trials into the comparative effectiveness of different vasoactive treatment strategies and within important specific patient subgroups.

Limitations

While the study benefits from a large cohort, its limitations should be acknowledged. First, the observational design

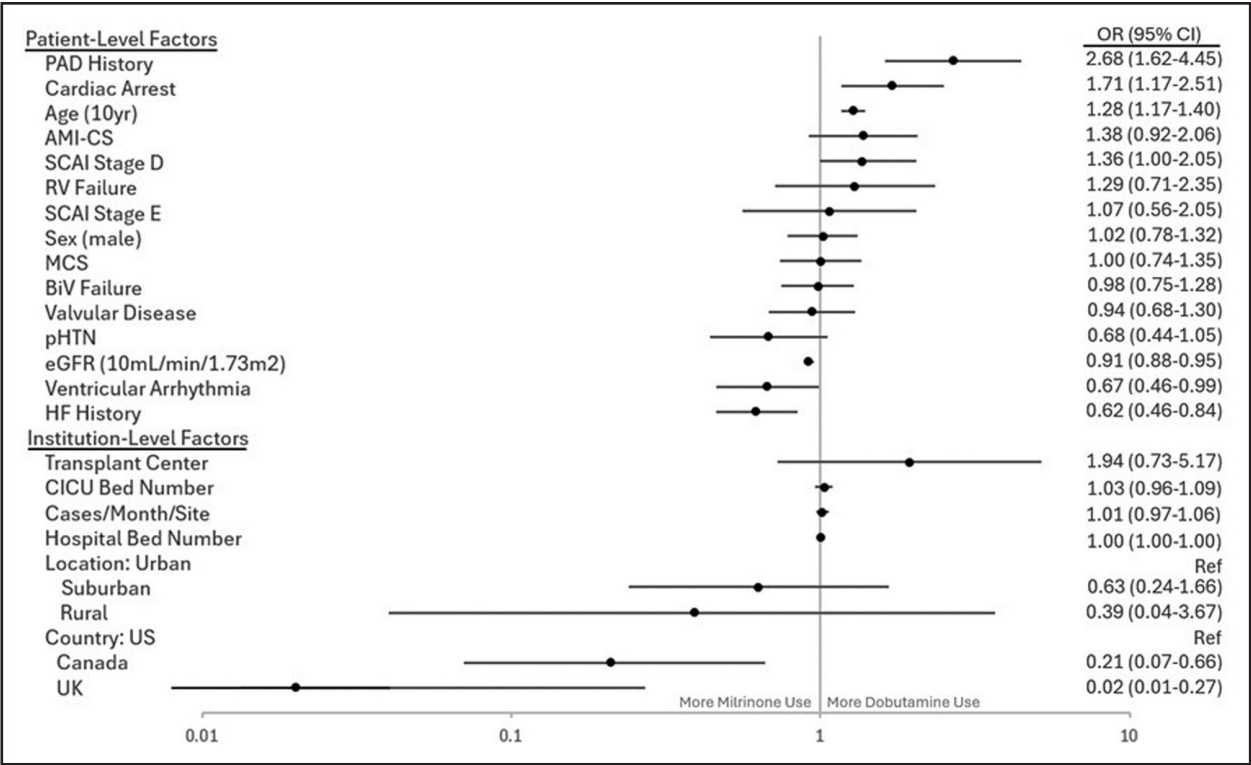


Figure 5. Patient- and institution-level factors associated with use of dobutamine vs milrinone treatment for patients in cardiogenic shock. Multivariable mixed-effect logistic modeling demonstrating odds ratio (OR) and 95% CI (95% CI) graphically displayed in log-scale. AMI-CS indicates acute myocardial infarction cardiogenic shock; BiV, biventricular; CICU, cardiac intensive care unit; eGFR, estimated glomerular filtration rate; HF, heart failure; LV, left ventricular; MCS, mechanical circulatory support; PAD, peripheral artery disease; pHTN, pulmonary hypertension; RV, right ventricular; SCAI, Society of Cardiovascular Angiography & Interventions; UK, United Kingdom; and US, United States.

precludes causal inference. Unmeasured confounders such as patient- and institutional-variables (eg, pharmacy-driven treatment protocols, ICU staffing models, ICU attending subspecialty makeup) not included in the analysis may contribute to the observed variability. Thus, we did not attempt to address the role of treatment variability on outcomes such as mortality or length of stay, which may yield misleading conclusions. In addition, this analysis is limited to the initial management and treatment selection of vasoactive agents within the first 24 hours of ICU admission. Further study is needed into the role of patient- and institution-level factors as well as pulmonary artery catheter and invasive hemodynamic data, on the timing, treatment sequencing, and evolution of CS care throughout the hospitalization. Levosimendan use is common in centers outside of the United States, and further evaluation into the variability of vasoactive treatment selections in centers that use levosimendan is needed. Finally, this large registry cohort derives from a select group of specialty CICU centers predominantly in urban areas across the United States, Canada, and the United Kingdom, with institutions and clinicians motivated for participation, and might not be generalizable to other institutions and clinical settings, including the Emergency Department, where vasoactive agents are typically initiated.

Conclusions

Overall, this study identifies substantial variability in vasoactive medication use for the treatment of CS and provides insights into the patient- and institution-level factors associated with these differences. These findings highlight the need for randomized trials comparing vasoactive treatments in patients with CS, as well as implementation studies to evaluate strategies for reducing variability and developing evidence-based guidelines that could enhance the consistency and quality of care for patients with CS.

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Received September 24, 2025; accepted April 13, 2026.

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Sources of Funding

None.

Disclosures

Dr Mathis has received research grants from the US National Institutes of Health (NIDDK) and Chiesi, USA, unrelated to the current work and paid to the University of Michigan. Dr Alviar has received research grants from Baxter Medical and the American College of Cardiology and consulting fees from Zoll, Johnson & Johnson, and Abbott. Dr Berg has received consulting fees from AstraZeneca, Pfizer, Mobility Bio, Inc, and Youngene Therapeutics, honoraria from the Medical Education Speakers Network (MESN), Metabolic Endocrine Education Foundation, Pri-Med, and USV Private Limited, and participates on clinical end point committees for studies sponsored by Beckman Coulter, Kowa Pharmaceuticals, Novo Nordisk, and Tosoh Biosciences. Dr Carnicelli receives research support from Acorai and has received speaking honoraria and travel support from Abiomed. Dr Daniels has served on clinical end point adjudication committees for Abbott, QuidelOrtho, and Tosoh, and has received consulting fees from Abbott, QuidelOrtho, Roche Diagnostics, and Siemens; and has received research grants from RCE and Roche. Dr Gage is a prior member of the Abiomed Speaker's Bureau and received an honorarium for Zoll Fellows' Education. Dr Katz has received speaker honoraria (modest) for Zoll, Abiomed, and AOP Health. Dr Proudfoot is funded by the Barts Charity and the Medical Research Council (Reference MR/W03011X/1). He has received institutional grants from J&J MedTech Heart Recovery and Getinge. Dr Solomon receives research support from the National Institutes of Health (NIH) Clinical Center intramural research funds. Dr Tomey has received fees for editorial services from McGraw-Hill Education, sponsored travel from the American College of Cardiology and the American Heart Association, and his spouse has received consulting fees from Medtronic. Dr Park, Patel, Berg, Bohula, and Morrow are members of the Thrombolysis in Myocardial Infarction Study Group, which has received institutional research grant support through Brigham and Women's Hospital from Abbott, Abiomed, Amgen, Anthos Therapeutics, ARCA Biopharma, Inc, AstraZeneca, Bayer HealthCare Pharmaceuticals, Inc, Daiichi-Sankyo, Eisai, 4TEEN4, Intarcia, Ionis Pharmaceuticals, Inc, Janssen Research and Development, LLC, MedImmune, Merck, Novartis, Pfizer, Quark Pharmaceuticals, Regeneron Pharmaceuticals, Inc, Roche, Siemens Healthcare Diagnostics, Inc, Softcell Medical Limited, and Zora Biosciences. Dr Morrow has received consulting fees from Abbott Laboratories, Merck, Novartis, Regeneron, and Roche. The views, information or content, and conclusions presented do not necessarily represent the official position or policy of, nor should any official endorsement be inferred on the part of, the Clinical Center, the NIH, or the Department of Health and Human Services. Dr Thompson is supported by NIH-National Heart, Lung, and Blood Institute (K08HL163328 and 1R01HL167524) and receives compensation as an editor for Merck Manuals, unrelated to the current work. The other authors report no conflicts.

Supplemental Material

Tables S1–S11

CCCTN Leadership and Investigators

REFERENCES

- Osman M, Syed M, Patibandla S, Sulaiman S, Khairi B, Shah MK, Bianco C, Balla S, Patel B. Fifteen-year trends in incidence of cardiogenic shock hospitalization and in-hospital mortality in the United States. *J Am Heart Assoc*. 2021;10:e021061. doi: 10.1161/JAHA.121.021061
- Malik A, Basu T, VanAken G, Aggarwal V, Lee R, Abdul-Aziz A, Birati EY, Basir MB, Nallamothu BK, Shore S. National trends for temporary mechanical circulatory support utilization in patients with cardiogenic shock from decompensated chronic heart failure: incidence, predictors, outcomes, and cost. *J Soc Cardiovasc Angiogr Interv*. 2023;2:101177. doi: 10.1016/j.jscv.2023.101177
- Parlow S, Fernando SM, Pugliese M, Qureshi D, Talarico R, Sterling LH, van Diepen S, Herridge MS, Price S, Brodie D, et al; LOTUS-ICU Research Group. Resource utilization and costs associated with cardiogenic shock complicating myocardial infarction: a population-based cohort study. *JACC Adv*. 2024;3:101047. doi: 10.1016/j.jaccadv.2024.101047
- Kapur NK, Kanwar M, Sinha SS, Thayer KL, Garan AR, Hernandez-Montfort J, Zhang Y, Li B, Baca P, Dieng F, et al. Criteria for defining stages of cardiogenic shock severity. *J Am Coll Cardiol*. 2022;80:185–198. doi: 10.1016/j.jacc.2022.04.049
- Lüsebrink E, Binzenhöfer L, Adamo M, Lorusso R, Mebazaa A, Morrow DA, Price S, Jentzer JC, Brodie D, Combes A, et al. Cardiogenic shock. *Lancet*. 2024;404:2006–2020. doi: 10.1016/S0140-6736(24)01818-X
- Bloom JE, Chan W, Kaye DM, Stub D. State of shock: contemporary vasopressor and inotrope use in cardiogenic shock. *J Am Heart Assoc*. 2023;12:e029787. doi: 10.1161/JAHA.123.029787
- Backer DD, Biston P, Devriendt J, Madl C, Chochrad D, Aldecoa C, Brasseur A, Defrance P, Gottignies P, Vincent J-L, et al. Comparison of dopamine and norepinephrine in the treatment of shock. *N Engl J Med*. 2010;362:779–789. doi: 10.1056/NEJMoa0907118
- Mathew R, Santo PD, Jung RG, Marbach JA, Hutson J, Simard T, Daniel Ramirez F, Harnett DT, Merdad A, Almufleh A, et al. Milrinone as compared with dobutamine in the treatment of cardiogenic shock. *N Engl J Med*. 2021;385:516–525. doi: 10.1056/NEJMoa2026845
- Levy B, Clere-Jehl R, Legras A, Morichau-Beauchant T, Leone M, Frederique G, Quenot J-P, Kimmoun A, Cariou A, Lassus J, et al; Collaborators. Epinephrine versus norepinephrine for cardiogenic shock after acute myocardial infarction. *J Am Coll Cardiol*. 2018;72:173–182. doi: 10.1016/j.jacc.2018.04.051
- Myburgh JA, Higgins A, Jovanovska A, Lipman J, Ramakrishnan N, Santamaria J; CAT Study Investigators. A comparison of epinephrine and norepinephrine in critically ill patients. *Intensive Care Med*. 2008;34:2226–2234. doi: 10.1007/s00134-008-1219-0
- Mebazaa A, Nieminen MS, Packer M, Cohen-Solal A, Kleber FX, Pocock SJ, Thakkar R, Padley RJ, Pöder P, Kivikko M; SURVIVE Investigators. Levosimendan vs dobutamine for patients with acute decompensated heart failure: the SURVIVE Randomized Trial. *JAMA*. 2007;297:1883–1891. doi: 10.1001/jama.297.17.1883
- van Diepen S, Katz JN, Albert NM, Henry TD, Jacobs AK, Kapur NK, Kilic A, Menon V, Ohman EM, Sweitzer NK, et al; American Heart Association Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; Council on Quality of Care and Outcomes Research; and Mission: Lifeline. Contemporary management of cardiogenic shock: a scientific statement from the American Heart Association. *Circulation*. 2017;136:e232–e268. doi: 10.1161/CIR.0000000000000525
- Heidenreich PA, Bozkurt B, Aguilar D, Allen LA, Byun JJ, Colvin MM, Deswal A, Drazner MH, Dunlay SM, Evers LR, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2022;79:e263–e421. doi: 10.1016/j.jacc.2021.12.012
- Sinha SS, Morrow DA, Kapur NK, Kataria R, Roswell RO. 2025 concise clinical guidance: an ACC expert consensus statement on the evaluation and management of cardiogenic shock. *J Am Coll Cardiol*. 2025;85:1618–1641. doi: 10.1016/j.jacc.2025.02.018
- Tavazzi G, Rossello X, Grand J, Gierlotka M, Sionis A, Ahrens I, Hassager C, Price S. Epidemiology, monitoring, and treatment strategy in cardiogenic shock. A multinational cross-sectional survey of ESC-acute cardiovascular care association research section. *Eur Heart J Acute Cardiovasc Care*. 2022;11:706–711. doi: 10.1093/ehjacc/zuac087

16. Mathis MR, Janda AM, Kheterpal S, Schonberger RB, Pagani FD, Engoren MC, Mentz GB, Shook DC, Muehlschlegel JD; Multicenter Perioperative Outcomes Group. Patient-, clinician-, and institution-level variation in inotrope use for cardiac surgery: a multicenter observational analysis. *Anesthesiology*. 2023;139:122–141. doi: 10.1097/ALN.0000000000004593
17. Berg DD, Barnett CF, Kenigsberg BB, Papolos A, Alviar CL, Baird-Zars VM, Barsness GW, Bohula EA, Brennan J, Burke JA, et al. Clinical practice patterns in temporary mechanical circulatory support for shock in the Critical Care Cardiology Trials Network (CCCTN) registry. *Circ Heart Fail*. 2019;12:e006635. doi: 10.1161/CIRCHEARTFAILURE.119.006635
18. Kadosh BS, Berg DD, Bohula EA, Park JG, Baird-Zars VM, Alviar C, Alzate J, Barnett CF, Barsness GW, Burke J, et al. Pulmonary artery catheter use and mortality in the cardiac intensive care unit. *JACC Heart Fail*. 2023;11(8, Part 1):903–914. doi: 10.1016/j.jchf.2023.04.007
19. Greenfield S, Nelson EC, Zubkoff M, Manning W, Rogers W, Kravitz RL, Keller A, Tarlov AR, Ware JE. Variations in resource utilization among medical specialties and systems of care. Results from the medical outcomes study. *JAMA*. 1992;267:1624–1630. doi:10.1001/jama.1992.03480120062034
20. McGlynn EA, Asch SM, Adams J, Keesey J, Hicks J, DeCristofaro A, Kerr EA. The quality of health care delivered to adults in the United States. *N Engl J Med*. 2003;348:2635–2645. doi: 10.1056/NEJMsa022615
21. Soni SM, Giboney P, Yee HF Jr. Development and implementation of expected practices to reduce inappropriate variations in clinical practice. *JAMA*. 2016;315:2163–2164. doi: 10.1001/jama.2016.4255
22. Sutherland K, Levesque JF. Unwarranted clinical variation in health care: definitions and proposal of an analytic framework. *J Eval Clin Pract*. 2020;26:687–696. doi: 10.1111/jep.13181
23. Miller PE, Huber K, Bohula EA, Krychtiuk KA, Pöss J, Roswell RO, Tavazzi G, Solomon MA, Kristensen SD, Morrow DA, et al. Research priorities in critical care cardiology. *JACC*. 2023;82:2329–2337. doi: 10.1016/j.jacc.2023.09.828
24. Bohula EA, Katz JN, van Diepen S, Alviar CL, Baird-Zars VM, Park J-G, Barnett CF, Bhattal G, Barsness GW, Burke JA, et al; Critical Care Cardiology Trials Network. Demographics, care patterns, and outcomes of patients admitted to cardiac intensive care units: the critical care cardiology trials network prospective North American Multicenter Registry of Cardiac Critical Illness. *JAMA Cardiol*. 2019;4:928–935. doi: 10.1001/jamacardio.2019.2467
25. Metkus TS, Baird-Zars VM, Alfonso CE, Alviar CL, Barnett CF, Barsness GW, Berg DD, Bertic M, Bohula EA, Burke J, et al. Critical Care Cardiology Trials Network (CCCTN): a cohort profile. *Eur Heart J Qual Care Clin Outcomes*. 2022;8:703–708. doi: 10.1093/ehjqcc/qcac055
26. Waksman R, Pahuja M, van Diepen S, Proudfoot AG, Morrow D, Spitzer E, Nichol G, Weisfeldt ML, Moscucci M, Lawler PR, et al. Standardized definitions for cardiogenic shock research and mechanical circulatory support devices: scientific expert panel from the Shock Academic Research Consortium (SHARC). *Circulation*. 2023;148:1113–1126. doi: 10.1161/CIRCULATIONAHA.123.064527
27. Patel SM, Berg DD, Bohula EA, Baird-Zars VM, Barsness GW, Chaudhry S-P, Chonde MD, Cooper HA, Ginder C, Jentzer JC, et al. Early serial assessment of aggregate vasoactive support and mortality in cardiogenic shock: insights from the critical care cardiology trials network registry. *Circ Heart Fail*. 2024;17:e011736. doi: 10.1161/CIRCHEARTFAILURE.124.011736
28. Nakagawa S, Schielzeth H. A general and simple method for obtaining R^2 from generalized linear mixed-effects models. *Methods Ecol Evol*. 2012;4:133–142. doi: 10.1111/j.2041-210x.2012.00261.x
29. Gaubert M, Laine M, Resseguier N, Aissaoui N, Puymirat E, Lemesle G, Michelet P, Hraiech S, Lévy B, Delmas C, et al. Hemodynamic profiles of cardiogenic shock depending on their etiology. *J Clin Med*. 2020;9:3384. doi: 10.3390/jcm9113384
30. Parlow S, Santo PD, Sterling LH, Goodliffe L, Motazedian P, Prosperi-Porta G, Morgan B, Koopman Z, Jung RG, Lepage-Ratte MF, et al. Inotrope versus placebo therapy in cardiogenic shock: rationale and study design of the CAPITAL DOREMI2 trial. *Am Heart J*. 2023;262:83–89. doi: 10.1016/j.ahj.2023.04.010
31. Di SP, Dehghan K, Mao B, Jung RG, Fadare D, Paydar J, Parlow S, Motazedian P, Prosperi-Porta G, Abdel-Razek O, et al. Milrinone vs dobutamine for the management of cardiogenic shock. *JACC Adv*. 2023;2:100393. doi: 10.1016/j.jacadv.2023.100393
32. Mathis MR, Mirizzi K, Burns CJ, Janda AM, Mentz G, Aaronson KD, Wu Z, Likosky DS, Pagani FD, Kheterpal S, et al. Clinician attitudes, opinions and practice patterns regarding inotrope use for cardiac surgery in the USA: a multicentre mixed methods study protocol. *BMJ Open*. 2025;15:e100306. doi: 10.1136/bmjopen-2025-100306